

AI-Assisted Product Risk & Business Performance Analytics

An end-to-end machine learning solution transforming raw transactional data into actionable portfolio risk intelligence for proactive business decision-making.



Business Goals



Early Risk Detection

Identify structurally weak or underperforming products before they impact profitability.



Revenue Exposure

Measure and monitor revenue exposure from high-risk SKUs across the portfolio.



Catalog Efficiency

Improve catalog efficiency and profitability through data-driven rationalization.



Executive Insights

Provide leadership with clear, actionable intelligence for strategic decisions.

Dataset Overview

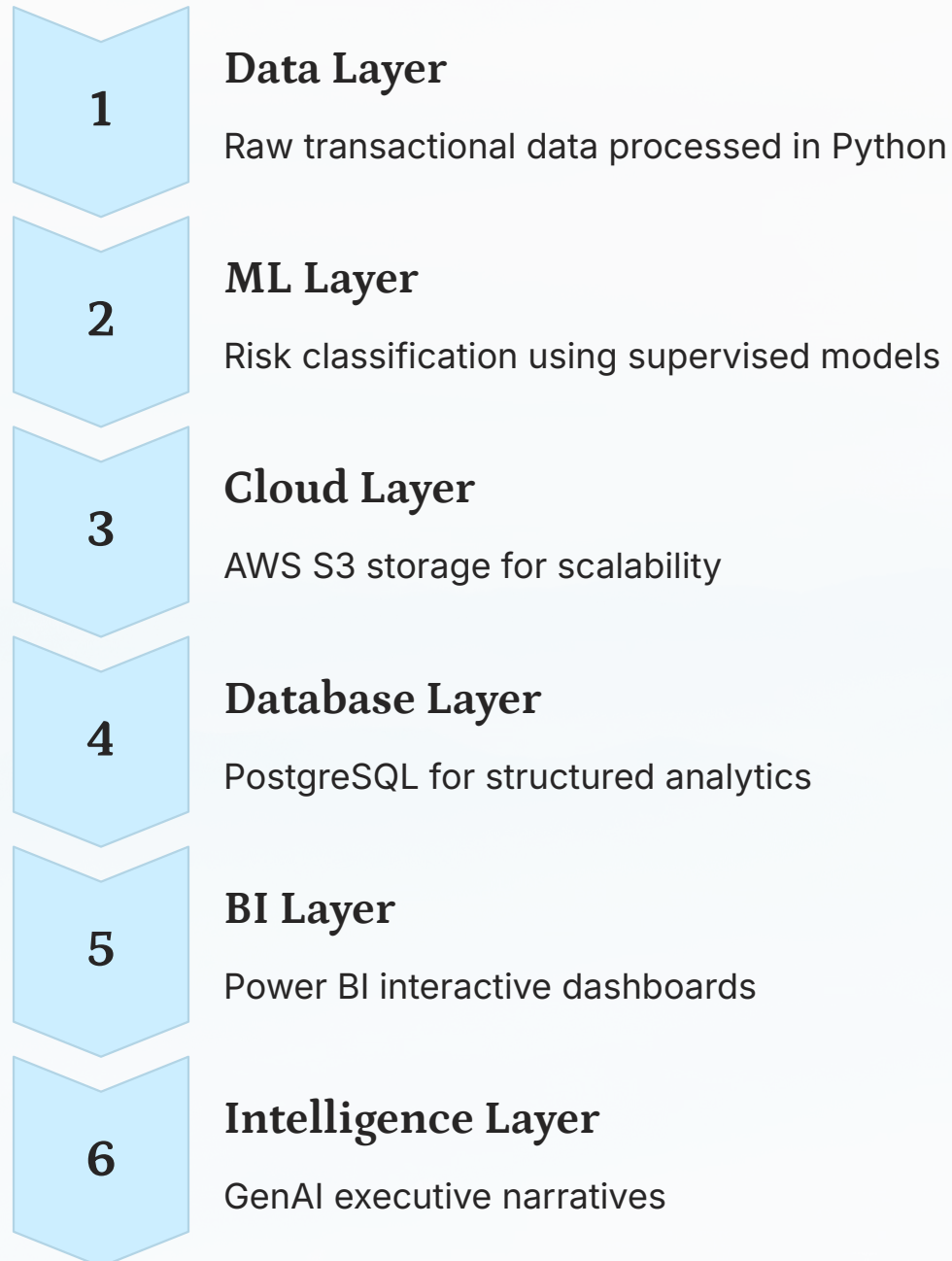
Data Foundation

The system analyzes **300K+ transactional rows** from online retail operations, covering products, customers, and orders.

- Revenue, units sold, and order frequency tracked
- Derived risk labels: High Risk / Low Risk
- Comprehensive product and customer entities



Cloud & Analytics Architecture



Machine Learning Design

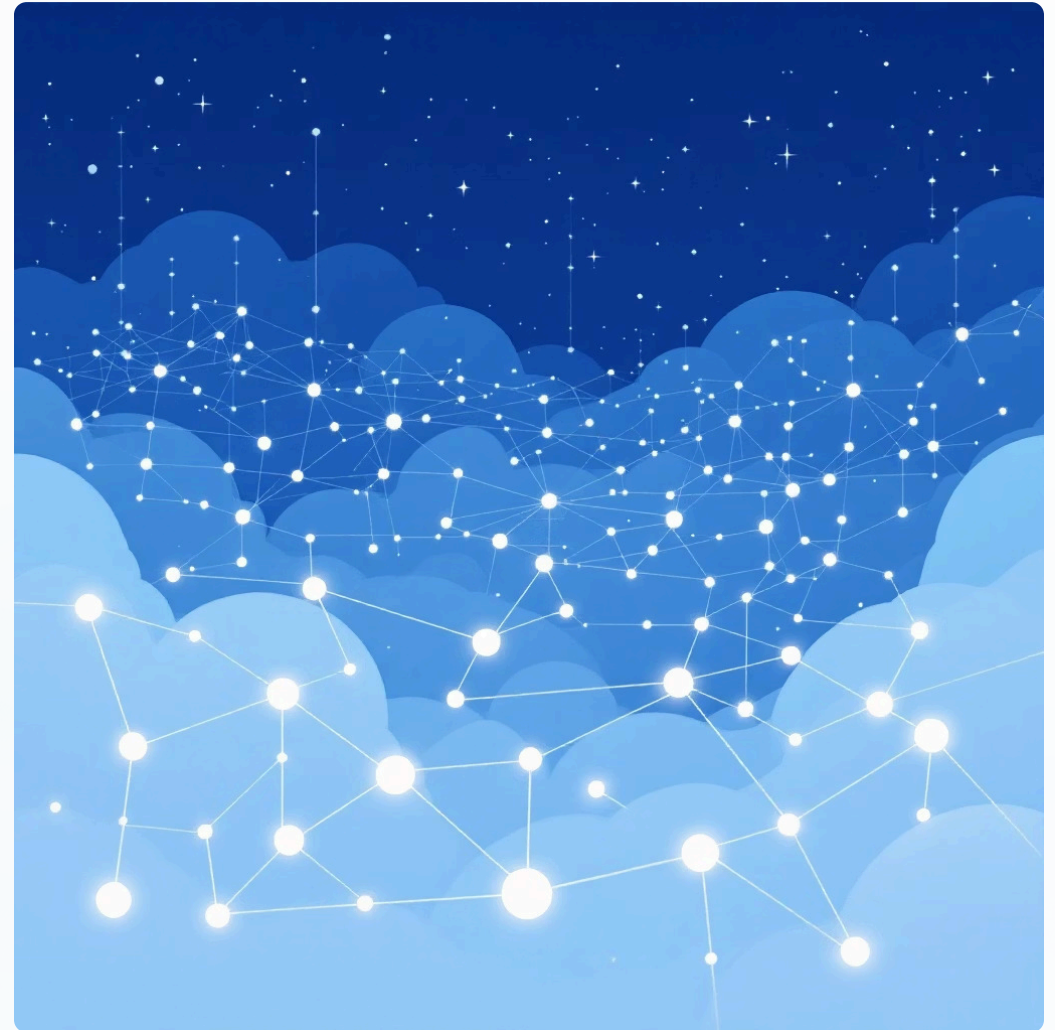
Problem Framing

Binary classification predicting **High Risk** or **Low Risk** products based on:

- Total Revenue
- Total Units Sold
- Order Count

Models Evaluated

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- Support Vector Machine (SVM)



Final Model: SVM

Selected for highest F1-Score, strong generalization, and stability across validation samples.

Critical Findings

52%

High Risk Products

Over half the catalog classified as structurally weak

8%

Revenue Contribution

High-risk products generate minimal revenue

3,194

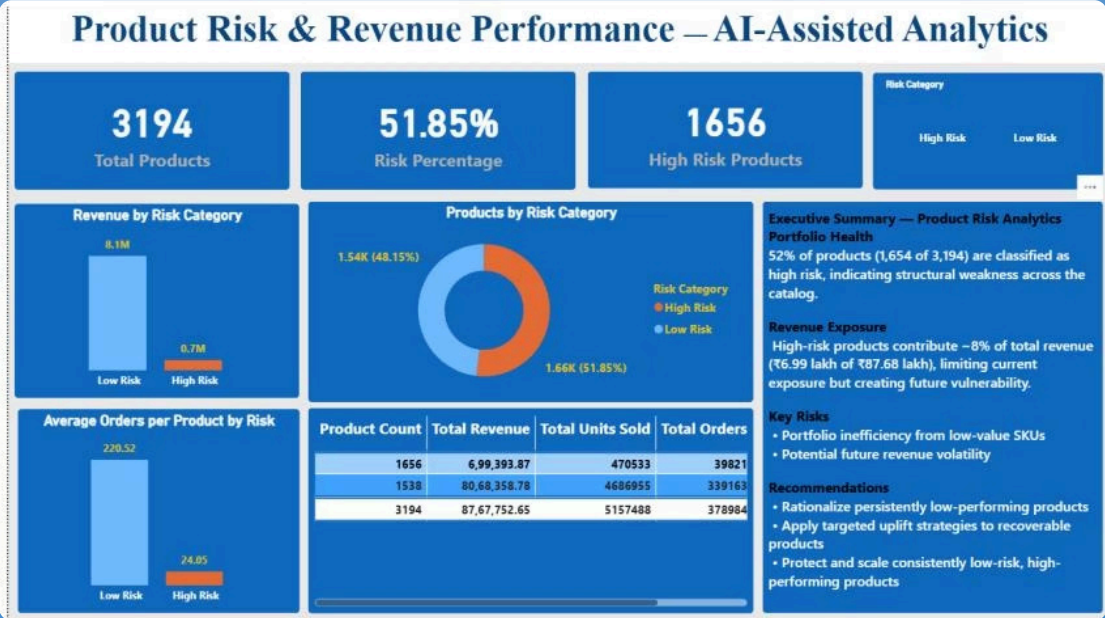
Total Products

Complete catalog analyzed for risk assessment

A large portion of the catalog generates low value but high operational overhead. Revenue concentration exists among a smaller subset of stable, low-risk products.

Power BI Dashboard

Interactive dashboard providing executive KPIs, risk distribution by category, revenue contribution analysis, high-risk SKU identification, and portfolio health indicators.



Executive Summary Insights

Portfolio Health

1,654 of 3,194 products (52%) classified as high risk, indicating structural weakness across the catalog.

Revenue Exposure

High-risk products contribute €6.99 lakh of €87.68 lakh total revenue (~8%), limiting current exposure but creating future vulnerability.

Performance Gap

Low-risk products average 220.52 orders vs. 24.05 for high-risk products—a 9x difference in performance.

An illustration of three business professionals (two men and one woman) sitting around a white table in a modern office setting. They are engaged in a meeting, with one person holding a tablet displaying a bar chart. The background features large windows and stylized blue plants.

Strategic Recommendations

01

Rationalize Low Performers

Remove persistently low-performing products to reduce operational overhead and improve catalog efficiency.

02

Apply Targeted Uplift

Implement recovery strategies for products with potential, focusing resources where they can drive impact.

03

Protect & Scale Winners

Invest in consistently low-risk, high-performing products to stabilize and grow revenue streams.

Business Impact & Future

Immediate Benefits

- Proactive portfolio optimization
- Reduced operational inefficiency
- Improved revenue focus on high-value products
- Faster executive decision-making

Scalable Foundation

The system provides infrastructure for ongoing portfolio optimization, risk management, revenue stabilization, and executive transparency.

